

SOFTWARE ENGINEERING – ASSESSMENT TOOL 1

NAME: Faraj Khaleel

REG NO: 192525134

COURSE CODE: CSA1007

PROBLEM NO: 36

TITLE: Autonomous Electric Vehicle Charging Network Management System

1. Project Vision & Stakeholder Identification

- **Project Vision:** To establish a distributed, secure, and intelligent cloud-enabled platform that automates electric vehicle charging network management, minimizes operational downtime, and ensures high availability and optimized resource utilization across multiple locations.
- **Project Objectives:** Collect real-time operational data, monitor performance continuously, detect anomalies automatically, predict failures using analytics, notify stakeholders instantly, generate comprehensive reports, and ensure high scalability and security.
- **Key Stakeholders & Roles:**
 - **Network Administrator:** Manages overall system configuration, user permissions, and network-wide infrastructure settings
 - **Maintenance Engineer:** Oversees hardware health, addresses predicted equipment risks, and resolves anomaly alerts.
 - **Operations Manager:** Monitors high-level system efficiency, oversees performance dashboards, and analyzes reports.
 - **EV Customer / End User:** Interacts with charging stations, authenticates sessions, and utilizes charging services.

2. Epics & User Stories

- **Epic 1: Real-Time Data Acquisition & Monitoring**

- User Story 1: As a Network Administrator, I want the system to continuously ingest IoT sensor data from charging stations, so that I can have real-time visibility into operational parameters like voltage and temperature.
- User Story 2: As an Operations Manager, I want a centralized dashboard showing live metrics, so that I can monitor system performance across multiple locations.

- **Epic 2: Anomaly Detection & AI Failure Prediction**

- User Story 1: As a Maintenance Engineer, I want the system to automatically flag abnormal events and operational anomalies, so that I can intervene before a failure occurs.
- User Story 2: As a Maintenance Engineer, I want AI/ML-driven risk and failure predictions, so that I can schedule proactive maintenance and reduce downtime.

- **Epic 3: Stakeholder Notification & Alerting Hub**

- User Story 1: As a Maintenance Engineer, I want to receive real-time automated alerts for critical system faults, so that I can respond immediately to emergencies.

3. Acceptance Criteria

- **For Data Acquisition User Story:**
 - **Given** IoT charging stations are actively online and operational, **When** telemetry data (voltage, temperature, connection status) is transmitted, **Then** the system must capture and store the data points in the database in real time.
- **For Anomaly Detection User Story:**
 - **Given** a charging station experiences abnormal voltage fluctuations, **When** the real-time processing engine evaluates the telemetry stream, **Then** an anomaly flag must be triggered and logged within 2 seconds.
- **For Real-Time Alerting User Story:**
 - **Given** a critical risk or anomaly is detected by the AI model, **When** the event is validated, **Then** the system must dispatch automated notifications to the designated maintenance engineer via the alert hub dashboard.

4. Product Backlog Prioritization & Sprint Planning

- **Product Backlog Prioritization (MoSCoW / Value-based)**
 1. High Priority: Real-Time Data Collection Engine & Ingestion Pipelines.
 2. High Priority: Centralized Dashboard Visualization.
 3. Medium Priority: Automated Anomaly Detection Module.
 4. Medium Priority: AI/ML Predictive Analytics & Risk Forecasting.
 5. Low Priority: Advanced Historical Report Generator.

- **Sprint 1 Backlog & Task Breakdown:**

- Selected User Stories: Data Ingestion and Centralized Dashboard.
- Actionable Tasks:
 - Set up cloud infrastructure (AWS/Azure) database connections.
 - Develop IoT protocol connection handlers (e.g., OCPP).
 - Code backend REST APIs for data collection using Python/Node.js.
 - Build frontend HTML/CSS/JavaScript dashboard interface.

5. Story Point Estimation, Sprint Goal & Presentation

- **Story Point Estimation (Fibonacci Scale)**

- Data Collection Pipeline: 8 Story Points (High complexity due to concurrent IoT streams and protocol mapping).
- Centralized Dashboard Interface: 5 Story Points (Standard frontend layout with real-time UI binding).
- Justification: Estimates are derived based on technical complexity, integration dependencies across distributed cloud nodes, and real-time synchronization requirements.

- **Sprint Goal:**

- Successfully establish the core data ingestion pipeline and functional live dashboard layout to ensure basic real-time operational visibility across charging stations.

- **Expected Sprint Deliverables:**

- Functional backend data intake module.
- Working prototype of the centralized dashboard displaying live simulated station metrics.
- Initial unit test scripts for data validation and pipeline reliability.